

AE 442/443 SENIOR DESIGN

SPACE LAUNCH VEHICLE DESIGN CHALLENGE

Request for Proposal (RFP)

Kairos Mars Launch Service

**University of Illinois Urbana-Champaign
Department of Aerospace Engineering**

Fall 2026 Senior Design Project

I. PURPOSE AND COMPETITION FRAMEWORK

The technical proposal and resulting launch vehicle design are the central products of this senior design project. Two student teams will act as competing Launch Vehicle Providers (LVPs) responding to a common customer need. Each team will develop an integrated space launch vehicle capable of delivering the complete Kairos customer payload onto a physically feasible Earth-to-Mars transfer trajectory.

The project is intentionally framed as a launch-service acquisition rather than an open-ended exercise to “design a rocket.” The customer payload and mission need are fixed. The vehicle architecture is not. Teams must translate the customer need into requirements, develop and evaluate competing vehicle concepts, select an architecture, mature it through formal design reviews, and demonstrate that the resulting system closes technically, programmatically, and economically.

Proposal Expectations

1. Demonstrate a thorough understanding of the RFP and Payload Interface Control Document (ICD) requirements.
2. Describe the technical approach used to satisfy each major mission and vehicle requirement, including the phasing and integration of subsystem work.
3. Identify critical technical and programmatic problem areas and provide sufficient analysis to permit engineering evaluation of the proposed solution.
4. Perform and document trade studies used to arrive at major vehicle architecture and subsystem decisions, including clear rationale for the selected design.
5. Describe the analytical, computational, and automated design tools used, including inputs, assumptions, limitations, verification, and validation where applicable.
6. Maintain traceability from customer requirements to vehicle-level and subsystem-level requirements and to the analyses or verification methods used to demonstrate compliance.

II. BASIS FOR FINAL EVALUATION

Consistent with the AIAA competition framework, evaluation will emphasize technical correctness, integrated launch vehicle design, requirements compliance, engineering decision quality, design maturity, originality, and practical feasibility rather than presentation polish alone. The strongest design is not necessarily the largest, most complex, or highest-performance launch vehicle. It is the vehicle that most convincingly satisfies the customer mission with appropriate margin while balancing performance, cost, risk, manufacturability, operability, and schedule.

Specific grading weights for design reviews, reports, analyses, individual contributions, and final deliverables will be defined in the course syllabus and review instructions. Project evaluation will prioritize: (1) technical correctness and depth; (2) integrated launch vehicle design; (3) systems engineering and requirements compliance; (4) quality of analysis and engineering trades; (5) practical feasibility; (6) technical communication; and (7) individual technical ownership and contribution.

III. REQUEST FOR PROPOSAL

Kairos Mars Launch Service

Background

Human and robotic activity around Mars will increasingly depend on reliable space-weather awareness and communications. Kairos was developed as a two-spacecraft heliophysics and communications architecture to support this need. Martian is intended for operations near the Sun-Mars L1 region and Helios near Sun-Mars L5. The spacecraft share a common bus architecture and carry their own propulsion for trajectory correction, Mars arrival, and final mission insertion. [1]

For purposes of this design challenge, the Kairos Program Office is preparing for a 2034 launch and is soliciting a new launch service. The customer does not prescribe a launch site, number of stages, propellant combination, engine cycle, vehicle diameter, recovery concept, or manufacturing approach. Those decisions belong to the Launch Vehicle Provider.

The customer is purchasing a capability: safely deliver both Kairos spacecraft from Earth to the required Trans-Mars Injection (TMI) condition and separate them onto a viable Mars-bound trajectory. The launch vehicle mission ends after successful TMI and spacecraft separation. The spacecraft assume responsibility for interplanetary cruise, trajectory correction, Mars approach, Mars capture, and final mission insertion.

Customer Payload

Parameter	Martian	Helios	RFP Treatment
Nominal spacecraft bus size	1.5 m cube	1.5 m cube	Customer source; excludes stowed deployable hardware
Wet mass + 5% margin	2,156.31 kg	2,156.31 kg	Customer source
Combined spacecraft mass		4,312.62 kg	Derived / baseline payload

The nominal 1.5 m bus dimensions do not represent the complete launch-stowed spacecraft envelope. Stowed deployable hardware, including the high-gain antenna and solar-array assemblies, extends beyond the central bus. Launch-vehicle accommodation requirements are therefore defined using conservative interface bounds in Appendix A.

The Kairos mission does not specify a launch-configuration center-of-mass, moment-of-inertia, or final payload-adaptor analysis. Accordingly, this RFP supplies conservative interface bounds in Appendix A rather than asking teams to reverse-engineer spacecraft information. [1]

Mission Need

The Launch Vehicle Provider shall design a new orbital launch system capable of placing the complete Kairos payload onto an Earth-escape trajectory that enables transfer to Mars during the 2034 launch opportunity. The original Kairos mission study estimated Earth-departure characteristic energy, C3, in the range of approximately 8–11 km²/s² depending on launch window. [1] This range shall be used for preliminary sizing; each team must determine and justify its final launch window and required C3.

IV. DESIGN REQUIREMENTS AND CONSTRAINTS

The following requirements establish the minimum mission and launch-vehicle capability expected of each proposal. They are intentionally written at a level that leaves meaningful design authority with the student teams. Derived subsystem requirements shall be developed by the teams and traced to these higher-level requirements.

Design Fidelity and Scope

Teams are expected to develop the launch vehicle itself to the highest level of technical fidelity achievable within the course. Manufacturing, ground systems, launch operations, test infrastructure, cost, and program planning shall be developed to a level sufficient to demonstrate system feasibility and identify major interfaces, requirements, resources, and risks; detailed launch-complex, factory, or ground-support-equipment design is not required.

Core launch vehicle disciplines including mission/trajectory, vehicle sizing and performance, propulsion, internal configuration, structures/materials, aerodynamics/loads, GNC, avionics, payload integration, and mass properties are expected to receive the greatest technical depth. Enabling functions such as separation, fairing, flight safety, verification/test planning, manufacturing, transportation, ground processing, and launch-site operations shall be developed at the fidelity necessary to demonstrate an integrated and credible vehicle.

A. Mission and Performance

LV-001	The launch vehicle shall deliver the complete Kairos baseline payload onto a physically feasible Earth-to-Mars transfer trajectory during the 2034 Mars launch opportunity.
LV-002	The baseline payload shall consist of Martian and Helios launched together on a single launch vehicle.
LV-003	The launch vehicle mission shall terminate following successful Trans-Mars Injection and separation of both customer spacecraft.
LV-004	For preliminary sizing, the Launch Vehicle Provider shall evaluate Earth-departure C3 values from 8 to 11 km ² /s ² .
LV-005	The Launch Vehicle Provider shall determine and justify the final launch date, launch window, departure geometry, C3, time of flight, and relevant trajectory margins.
LV-006	The launch vehicle shall demonstrate positive performance margin for the selected mission using vehicle-specific propulsion, aerodynamic, gravity, steering, and operational losses.

B. Launch Site and Mission Architecture

LV-007	The launch site shall be selected by the Launch Vehicle Provider and justified using trajectory performance, launch azimuth, range safety, overflight, infrastructure, weather, environmental constraints, logistics, regulatory considerations, and cost.
LV-008	The vehicle may use direct injection, a parking-orbit strategy, or another technically credible departure architecture provided the team demonstrates mission closure and operational feasibility.
LV-009	The Launch Vehicle Provider shall identify primary and backup launch opportunities and assess sensitivity to launch delay within the 2034 Mars opportunity.

C. Payload Accommodation

LV-010	The launch vehicle shall accommodate a customer spacecraft mass of 4,313 kg, excluding launch-vehicle-provided payload adapters, dispensers, separation systems, and other launch-specific hardware.
LV-011	The launch vehicle shall satisfy the mechanical, mass-property, structural, and operational interfaces defined in Appendix A, Kairos Payload Interface Control Document.
LV-012	The Launch Vehicle Provider shall design and include all required payload support, adapter, load-transfer, restraint, and separation hardware in the launch vehicle mass budget.
LV-013	The payload fairing shall accommodate the complete launch-stowed Kairos configuration, including stowed deployable hardware, within the payload accommodation envelope defined in Appendix A, with appropriate structural, dynamic, integration, and separation clearances.
LV-014	The Launch Vehicle Provider shall define the spacecraft arrangement and deployment sequence and demonstrate that the selected architecture prevents recontact.

D. Vehicle Design and Integration

LV-015	The launch vehicle architecture shall be supported by quantitative trades addressing staging, propulsion, structural architecture, mission performance, cost, risk, manufacturability, and operations.
LV-016	The team shall maintain an integrated vehicle mass-properties model and mass budget from SRR through CDR, including appropriate subsystem and system margin.
LV-017	The team shall establish vehicle-level and subsystem-level requirements with traceability to the RFP and shall maintain a requirements compliance matrix.
LV-018	Major vehicle interfaces shall be defined and controlled, including propulsion-to-structure, stage-to-stage, avionics-to-GNC, payload-to-vehicle, and ground-to-flight interfaces.
LV-019	Integrated Vehicle Configuration: The Launch Vehicle Provider shall develop an integrated internal and external vehicle configuration demonstrating the physical accommodation and interfaces of major flight systems.

The CAD/configuration per LV-019 will show, as applicable:

- Propellant tanks and domes
- Engines/motors
- Thrust structure
- Feedlines and major valves
- Pressurization systems
- Intertank/interstage structure
- Avionics bays
- Batteries/power
- Flight computers
- GNC sensors
- Telemetry hardware
- Flight termination hardware
- Payload adapter/dispenser
- Payloads
- Fairing
- Separation planes and hardware
- Structural load paths

Subsystems shall not be represented solely as external vehicle allocations or conceptual blocks. Teams shall demonstrate that major systems physically fit within the vehicle and that their interfaces, access, load paths, integration, assembly, and servicing have been considered. By CDR, each team shall provide an integrated longitudinal section or equivalent cutaway configuration showing major internal components, dimensions,

separation planes, propellant volumes, payload accommodation, and primary structural load paths. The configuration shall be tied to the team's component-to-subsystem-to-stage-to-vehicle mass rollout.

E. Propulsion, Flight, and Control

The proposal shall provide sufficient propulsion and flight-system definition to support engineering evaluation. At a minimum, teams shall address:

- Propellant selection, propulsion cycle or motor architecture, engine count, thrust, specific impulse, mixture ratio where applicable, chamber pressure, throttling/restart needs, and stage performance.
- Tankage, feed systems, pressurization, propellant management, thermal considerations, and engine-to-stage integration.
- Ascent guidance, navigation, attitude control, staging control, TMI targeting, and control authority across the flight envelope.
- Flight computer, sensors, telemetry, tracking, command, power, redundancy, fault management, and a credible flight safety/termination concept.

F. Structures, Aerodynamics, Loads, and Environments

LV-020	The launch vehicle shall demonstrate positive structural margin for all governing load cases using appropriate factors of safety and clearly identified assumptions.
LV-021	The team shall identify and evaluate governing load cases including, as applicable, liftoff, maximum dynamic pressure, maximum aerodynamic loading, maximum axial acceleration, staging, engine shutdown/start, fairing separation, TMI, and payload separation.
LV-022	The launch vehicle shall maintain the customer spacecraft within the acceleration and dynamic compatibility limits defined in Appendix A.
LV-023	The team shall characterize aerodynamic performance and flight loads at a fidelity appropriate to the design maturity and demonstrate how aerodynamic analysis feeds trajectory, GNC, loads, and structural sizing.

G. Ground, Manufacturing, Test, and Operations

A technically capable vehicle that cannot be manufactured, tested, processed, or launched on the proposed schedule is not a viable solution. The design shall therefore include a credible concept for:

- Vehicle manufacturing and assembly, including major materials, joining methods, inspections, and producibility constraints.
- Component, subsystem, stage, and integrated vehicle development and qualification testing.
- Transportation, launch-site integration, payload processing, propellant loading, countdown, launch, scrub/abort response, and post-flight operations.
- Recovery and refurbishment when reuse is proposed.

Teams are not expected to develop detailed ground-support equipment, launch-complex, or manufacturing-facility designs. Instead, the Launch Vehicle Provider shall demonstrate that the proposed vehicle can credibly be manufactured, integrated, tested, transported, processed, and launched. These topics shall be addressed at conceptual/architectural fidelity with sufficient engineering rationale to identify major needs, interfaces, constraints, and risks. Student effort should remain centered on the launch vehicle itself.

Manufacturing

- Major manufacturing approach
- Assembly sequence
- Materials/processes
- Major tooling/facility needs
- Inspection/quality approach

Test

- Component → subsystem → stage → integrated vehicle
- Development vs qualification vs acceptance testing
- Major facilities needed
- Key verification events

Operations

- Transportation
- Vehicle integration
- Payload integration
- Rollout/transport to pad
- Propellant loading
- Countdown
- Launch
- Scrub/recycle
- Recovery, if applicable

LV-024	Launch Site Interface: The Launch Vehicle Provider shall define the major interfaces and services required from the selected launch site.
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Areas to consider include:

- Vehicle support/hold-down
- Launch mount interface
- Propellant commodities
- Fill/drain connections
- Pressurization/purge gases
- Electrical power
- Data/communications
- Umbilicals
- Environmental conditioning
- Ignition interface
- Flame/exhaust management
- Water/deluge if required
- Vehicle access
- Payload access
- Range safety
- Lightning/weather protection
- Emergency safing

H. Cost, Schedule, and Risk

LV-025	The Launch Vehicle Provider shall develop a credible estimate of non-recurring development cost and recurring launch cost using a clearly stated methodology and assumptions.
LV-026	The proposed program shall support a target first launch in 2034. Program formulation and architecture maturation begin in 2026, with a credible Design, Development, Test, and Evaluation (DDT&E) program progressing through first flight.
LV-027	The team shall identify major technical, cost, schedule, safety, and programmatic risks and shall propose mitigation or retirement plans appropriate to the design maturity.
LV-028	Lower cost shall not compensate for failure to satisfy the baseline mission; however, cost and commercial viability will be considered when comparing technically compliant designs.

I. Optional Rideshare Capability

After the baseline Kairos mission is technically closed, the Launch Vehicle Provider may propose additional rideshare capability as a commercial enhancement. Rideshare capability is optional, shall not drive the baseline architecture unless justified by a documented trade, and shall not be used to justify a vehicle that otherwise fails the Kairos mission.

LV-029	Any claimed rideshare capability shall identify available mass, volume, interface, destination compatibility, separation approach, impact to the baseline mission, incremental cost, and expected customer value.
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V. DESIGN REVIEWS AND PROGRAM MATURATION

The design shall mature through three formal customer-style design reviews followed by a final customer-facing provider package and technical defense. Detailed entrance, success, and exit criteria will be issued separately. The reviews are intended to force progressive system convergence rather than create independent presentation events.

Review	Primary Question	Expected Maturity
Combined SRR / System Definition Review	Do we understand the mission, and have we selected a credible system architecture?	Requirements baseline, ConOps, launch site/trajectory trades, architecture trades, preliminary staging/propulsion/configuration, mass budget, risks
PDR	Does the integrated preliminary vehicle design close?	Internal configuration, trajectory closure, propulsion sizing, tanks, structures, aero, loads, GNC, avionics, payload integration, preliminary ground interfaces, cost/schedule
CDR	Is the vehicle sufficiently defined and analyzed to support detailed development?	Mature integrated design, internal layout, interfaces, structural/load results, propulsion definition, verification approach, DDT&E plan, operations concept
Final Launch Vehicle Provider Package	Would a customer select this launch service?	Final customer-facing Launch Vehicle Provider document, ≥24 in engineering prototype, and final technical defense.

VI. DELIVERABLES

The design reviews and final deliverables shall communicate a complete, integrated launch vehicle design. Review-specific content and grading will be defined separately; the items below establish the expected content of the final Launch Vehicle Provider Package and supporting engineering products.

1. **Requirements Definition** – Define mission, vehicle, and subsystem requirements; identify parent requirements; maintain traceability; and demonstrate understanding of the RFP and ICD.
2. **Concept of Operations** – Describe all mission phases from manufacturing and ground integration through launch, ascent, staging, TMI, payload separation, and post-flight/recovery operations where applicable.
3. **Architecture and Trade Studies** – Document major vehicle and subsystem trades. Explain metrics, weighting when appropriate, assumptions, pros/cons, risk, cost, and rationale for final decisions.
4. **Integrated Launch Vehicle Design** – Present the complete vehicle architecture and subsystem designs, including propulsion, trajectory/GNC, avionics, aerodynamics, structures/materials, loads/analysis, payload integration, operations, and interfaces.
5. **Analysis Methods and Design Tools** – Describe key computational or analytical tools, model inputs and outputs, assumptions, verification/validation, convergence or sensitivity checks, and limitations.
6. **Mass, Performance, and Margin Summary** – Provide integrated mass properties, stage mass fractions, propellant budgets, performance closure, trajectory results, and clearly labeled margin at the vehicle and subsystem level.
7. **Cost and Business Case** – Provide development and recurring launch cost estimates, cost methodology, major WBS elements, infrastructure assumptions, and optional rideshare value where proposed.
8. **Development and Test Schedule** – Provide an integrated 2026–2034 program schedule beginning with formulation and architecture maturation in 2026 and progressing through DDT&E, hardware development, manufacturing, qualification, stage/integrated testing, launch-site readiness, and first flight in 2034.
9. **Risk and Verification** – Identify major risks, mitigation/retirement plans, verification methods, planned evidence, and status of compliance with major requirements.
10. **Final Launch Vehicle Provider Package** – Produce a professional customer-facing package that allows an external reviewer to understand what the vehicle is, what it can do, how it will be developed and operated, what it will cost, and why the customer should select it.
11. **Physical Launch Vehicle Prototype** – Each team shall produce a physical 3D-printed engineering model of its final launch vehicle configuration. The assembled vehicle shall be no less than 24 inches (610 mm) in overall height. The prototype shall communicate both the external configuration and major internal system arrangement and shall identify, as applicable, stages, tanks, propulsion systems, avionics, payload accommodation, interstages, fairing, and major separation planes. Teams may use removable fairings or skins, cutaway sections, separable stages, open sections, or other approaches that clearly communicate the

integrated design. The prototype is an engineering communication model; it is not required to be a flight-capable model rocket.

12. **Integrated Vehicle Configuration Drawing** – By CDR and in the final provider package, each team shall provide a dimensioned longitudinal section, cutaway, or equivalent integrated configuration drawing showing the major internal arrangement, interfaces, separation planes, payload accommodation, propellant volumes, and primary load paths. The drawing shall be consistent with the final CAD model, mass budget, and physical prototype.

VII. DESIGN FREEDOM

Except where constrained by the RFP or Payload ICD, the Launch Vehicle Provider retains design authority. Teams are expected to make and defend decisions rather than seek a prescribed “correct” launch vehicle.

- Number of stages and staging architecture
- Propellant combination and propulsion cycle
- Engine selection, adaptation, or new engine concept
- Vehicle diameter, length, tank layout, and structural architecture
- Launch site, ascent profile, parking-orbit or direct-injection strategy
- Fairing and payload support architecture
- GNC, avionics, telemetry, and flight software architecture
- Materials and manufacturing approach
- Expendable, partially reusable, or reusable architecture
- Ground operations and launch processing approach
- Optional rideshare strategy

Higher-Fidelity Analysis, Test, and Resource Planning. Teams are encouraged to identify critical design questions that would materially benefit from higher-fidelity computational analysis, experimental testing, prototyping, or laboratory evaluation. Examples include CFD, FEA, coupled loads, modal/buckling analysis, trajectory optimization, Monte Carlo dispersion analysis, separation dynamics, GNC simulation, thermal analysis, aerodynamic testing, or focused component tests.

Proposed work should answer a specific design question, reduce uncertainty, or retire a meaningful risk rather than exist solely to satisfy a test requirement. Teams requiring university laboratory resources, specialized test equipment, or high-performance computing/cluster resources shall identify those needs early and coordinate them with the instructional team. Resource availability is subject to technical merit, safety review, facility availability, schedule, and instructor approval.

VIII. SUPPORTING DATA AND GOVERNING DOCUMENTS

The following documents form the initial technical basis for the project. When conflicts exist, the AE 442/443 RFP and the latest issued Payload ICD govern the launch-vehicle design challenge.

7. Kairos spacecraft design report, University of Illinois Urbana-Champaign, 2026. Customer spacecraft architecture, mass, structural assumptions, mission analysis, and preliminary launch analysis. [1]
8. [2025–2026 AIAA Undergraduate Team Space Design Competition RFP](#). Used as a reference for RFP organization, proposal expectations, requirements framing, and deliverable structure. [2]
9. Course lectures, assigned launch-vehicle design references, applicable launch-provider user guides, and instructor-issued review criteria.
10. Appendix A to this RFP: Preliminary Kairos Payload Interface Control Document.

APPENDIX A — KAIROS PAYLOAD INTERFACE CONTROL DOCUMENT

This section establishes the controlled interface between the customer spacecraft and the student-designed launch vehicle. Values taken directly from the Kairos proposed mission are identified as [KAIROS]. Values calculated from source information are [DERIVED]. Course-defined requirements are [RFP]. Conservative engineering envelopes are [BOUND], and simplifying assumptions are [ASSUMPTION].

A1. Payload Mass and Geometry

[KAIROS] Each spacecraft uses a nominal 1.5 m × 1.5 m × 1.5 m central bus. The complete launch-stowed spacecraft includes deployable hardware extending beyond the nominal bus dimensions.

[KAIROS] Each spacecraft has a wet mass of 2,156.31 kg including the customer's stated 5% mass margin.

[DERIVED] The combined baseline spacecraft mass is 4,312.62 kg; use 4,313 kg for RFP compliance.

[BOUND] For launch-vehicle accommodation, each spacecraft shall be treated as occupying a maximum 3.0 m × 3.0 m × 2.0 m launch-stowed bounding envelope. This course-defined envelope conservatively accommodates the central spacecraft bus and stowed deployable hardware and does not represent the exact spacecraft outer mold line.

[RFP] Payload adapter, support, dispenser, restraint, and separation hardware are launch-vehicle hardware and are not included in the 4,313 kg customer mass.

[RFP] The Launch Vehicle Provider may select the orientation of each spacecraft within the payload fairing and may vertically stack, laterally arrange, or otherwise package the two spacecraft, provided all payload accommodation, structural, separation, and mission requirements are satisfied.

A2. Payload Configuration and Mass Properties

[KAIROS] No complete final launch-configuration center-of-mass or moment-of-inertia calculations.

[BOUND] The CG of each spacecraft in its launch-stowed configuration shall be assumed to lie within ±0.25 m in X, Y, and Z of the nominal spacecraft centerline and geometric center of the central bus. The resulting CG location shall remain within the defined launch-stowed payload envelope.

[BOUND] For preliminary coupled-dynamics, payload-adapter, and control analysis, each spacecraft in its launch-stowed configuration shall be assumed not to exceed the following principal moments of inertia:

$$I_{xx} \leq 2,500 \text{ kg m}^2$$

$$I_{yy} \leq 2,500 \text{ kg m}^2$$

$$I_{zz} \leq 3,500 \text{ kg m}^2$$

These values are conservative customer-defined bounds based on the launch-stowed accommodation envelope and are not intended to represent the exact spacecraft mass distribution.

A3. Structural and Dynamic Interface

[KAIROS] The spacecraft were analyzed against a 6 g axial and 2 g lateral launch load condition, with a first-mode requirement greater than 25 Hz.

[RFP] The 6.0 g axial and 2.0 g lateral conditions shall be treated as the preliminary spacecraft quasi-static structural load envelope for launch-vehicle integration. The Launch Vehicle Provider shall demonstrate that predicted spacecraft-interface accelerations and applicable launch load cases are compatible with this envelope.

[RFP] The launch vehicle shall demonstrate structural/dynamic compatibility with a spacecraft fundamental mode no lower than 25 Hz.

[ASSUMPTION] Detailed random-vibration, acoustic, and shock qualification spectra are not provided by the Kairos mission team and will be issued or bounded separately if required for later design maturation.

[RFP] Teams shall nevertheless characterize predicted payload vibration, acoustic, and shock environments and identify any unresolved compatibility risk.

A4. Payload Attach and Separation

[KAIROS] Approximately 400 kg for an adapter/stacking architecture in early launch study. Final payload attach fitting and adapter analysis remained to be completed.

[RFP] The Launch Vehicle Provider shall design the payload attach, load-transfer, support, and deployment architecture for both spacecraft.

[RFP] The spacecraft may be vertically stacked, side-by-side, or accommodated using another credible architecture; the RFP does not prescribe the configuration.

[RFP] Martian and Helios shall be independently and positively separated after successful TMI.

[RFP] The provider shall determine and justify separation sequence, separation velocity, tip-off rate, orientation, timing, and recontact clearance.

A5. Thermal, Contamination, and Electrical Interface

[KAIROS] Thermal analysis identifies a nominal operating range of approximately 10°C to 40°C. No complete launch-fairing qualification envelope.

[ASSUMPTION] Both spacecraft are electrically self-sufficient during launch unless a team proposes an alternate interface.

[RFP] The launch vehicle shall provide an environmentally controlled payload volume during ground processing and ascent sufficient to avoid customer thermal-limit violations.

[RFP] The launch vehicle shall identify contamination sources associated with propulsion, venting, fairing separation, pyrotechnic or non-pyrotechnic separation, and ground processing.

[RFP] The launch vehicle shall provide discrete command capability for spacecraft separation and appropriate separation-event telemetry.

A6. Preliminary Interface Summary

Interface Item	Baseline / Bound	Classification
Customer payload	Martian + Helios	KAIROS
Total customer mass	4,313 kg	DERIVED / RFP
Individual nominal central bus	1.5 m cube	KAIROS
Individual launch-stowed envelope	3.0 m x 3.0 x 2.0 m	BOUND
Launch-stowed CG uncertainty	±0.25 m XYZ	BOUND
Launch-stowed principal MOI	$I_{xx} \leq 2,500 \text{ kg m}^2$ $I_{yy} \leq 2,500 \text{ kg m}^2$ $I_{zz} \leq 3,500 \text{ kg m}^2$	BOUND
Max axial acceleration	6.0 g	KAIROS / RFP
Max lateral acceleration	2.0 g	KAIROS / RFP
Spacecraft first-mode floor	25 Hz	KAIROS / RFP
Payload arrangement	Provider selected	RFP
Payload adapter / separation	Provider designed	RFP

REFERENCES

- [1] University of Illinois Urbana-Champaign student design team, “Kairos” spacecraft design report, 2026. Primary customer source for spacecraft architecture, mass properties status, structural load assumptions, mission trajectory analysis, and preliminary launch analysis.
- [2] American Institute of Aeronautics and Astronautics, “2025–2026 AIAA Undergraduate Team Space Design Competition,” 2025. https://aiaa.org/wp-content/uploads/2025/08/2025-2026-Undergraduate-Team_Space-Design-Competition.pdf